

Empiricist

WEIGHT 3 · Nykolas Rekasius · LONGEST / DENSEST

Axes M/N/K · $Y=1-S$ power laws · Fig.5 fits · P1-P12 · RC0-RC5 · GPU honesty · reproduce

Claim under measurement

Single-task Diffusion Policy OOD success on new envs/objects is a power-law function of diversity counts (M, N, pairs) — NOT of raw demo count past modest $K \approx 50$.

Course success = directional mini-N rankings · do NOT claim Tab.1 ~90% from smoke

Shared spine — paste into every deck

Empiricist owns the numbers; others carry the tattoo

1 Diversity $\gg$ demo depth. Power laws on #envs / #objects / #pairs; demos plateau $\rightarrow K \approx 50$.

2 Recipe: 32 env-object pairs $\times$ ~ 50 demos $\rightarrow \sim 90\%$ zero-shot on novel env+object.

3 Opposite “just collect more demos.” Matched-total still favors diversity (App. G.2).

4 Must cite: Fig.5 exponents · Tab.1 SR 85–92.5% · ViT scales, U-Net doesn't · frozen/LfS fail.

5 Motion: DP chunks + ACT temporal ensemble @5 Hz Franka; action horizon 16; obs 2–3.

Formal axes — $M / N / K \cdot Y = 1 - S$

\$3 · power-law form CLEAR · 3-param $Y \propto$ not fit (only 6 points)

M

training environments

N

training objects (same cat.)

K

demos per env-object pair

S

normalized score on UNSEEN

Y

optimality gap = $1 - S$

X

$\in \{N, M, \text{\#pairs}\}$

$$Y = \beta \cdot X^\alpha \rightarrow \log Y = \alpha \log X + \log \beta$$

Fit on 100% demo curves only · no task/lang axis · nested larger $\supset$ smaller · final ckpt not best-val

Fig. 5 — power-law log-log fits

Six panels · optimality gap vs objects / envs / pairs · 100% demos

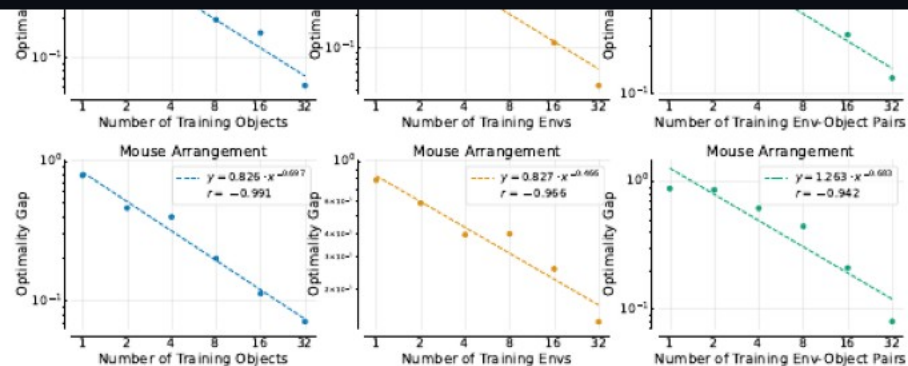


Figure 5: **Power-law relationship.** Dashed lines represent power-law fits, with the equations provided in the legend. All axes are shown on a logarithmic scale. The correlation coefficient (Pearson's r) indicates a power-law relationship between the generalization ability and the number of objects, environments, and environment-object pairs. See Appendix G.1 for data scaling laws on MSE.

ing further emphasizes that expanding the diversity of environments and objects is more effective than merely increasing the number of demonstrations for each individual environment or object.

4.2 POWER-LAW FITTING AND QUANTITATIVE ANALYSIS

We next explore whether our experimental results follow power-law scaling laws, as seen in other domains. Specifically, if two variables Y and X satisfy the relation $Y = \beta \cdot X^{\alpha}$, they exhibit a power-law relationship. Applying a logarithmic transformation to both Y and X reveals a linear relationship: $\log(Y) = \alpha \log(X) + \log(\beta)$. In our context, Y represents the optimality gap, defined as the deviation from the maximum score (i.e., $1 - \text{Normalized Score}$), while X can denote the number of environments, objects, or demonstrations. Using data from our previous experiments with a 100% fraction of demonstrations, we fit a linear model to the log-transformed data, as shown in Fig. 5. Based on all the results, we summarize the following data scaling laws:

¹Although we recognize the irreducible errors Y_{∞} associated with scaling the data alone, fitting a three-parameter model $Y = \beta X^{\alpha} + Y_{\infty}$ does not seem statistically justified given that we have only 6 data points.

Fig. 5 fits — all six equations + r

Re-fit from App. G.3 raw · Mouse 0.99 $\Rightarrow$ 1,191 pairs

Task	Axis X	Fit $Y=\beta X^\alpha$	r
Pour Water	# objects	$y = 0.825 \cdot x^{\{-0.703\}}$	0.987
Pour Water	# envs	$y = 1.180 \cdot x^{\{-0.844\}}$	0.960
Pour Water	# pairs	$y = 1.068 \cdot x^{\{-0.579\}}$	0.980
Mouse Arr.	# objects	$y = 0.826 \cdot x^{\{-0.697\}}$	0.991
Mouse Arr.	# envs	$y = 0.827 \cdot x^{\{-0.466\}}$	0.966
Mouse Arr.	# pairs	$y = 1.263 \cdot x^{\{-0.683\}}$	0.942

Demo-count axis: NO clear power law (Pour $r=-0.62$, Mouse $r=-0.79$) · plateaus · do not fit Y_∞ with $n=6$

Reproduce checklist

UMI collect → DP+DINOv2 → Franka eval → fit · project site + HF

- 1 Collect** UMI gripper + GoPro; SLAM filter (~90% valid); randomize init/object pose every demo
- 2 Train** DP CNN U-Net + DDIM-16; DINOv2 ViT-L/14 full FT; BF16; nested subsets; final ckpt
- 3 Deploy** Franka + WSG-50 + Hero10 wrist + 4090; 5 Hz; ACT TE on (8 steps, -0.01)
- 4 Eval** 8 held-out envs/objs × 5 trials; blind shuffle; step scores 0-3 → normalize; 95% CI
- 5 Fit** $Y=1-S$; OLS $\log Y \sim \log X$; report β, α, r ; skip Y_∞ ; verify Mouse 0.99 → ~1191
- 6 Recipe** New task: 32 pairs × 50 demos × 1 obj/env → expect ~90% SR if difficulty $\approx$ paper

Ablation shortlist P1-P12

AUTHORITATIVE: data-scaling-laws-2410.18647-ablations.md · min P1+P2+P5→P4→P8/P10→P11

P1	#envs M	Pour 0.144→0.956	RC1 PRIMARY
P3	#pairs	Pour 0.050→0.875	Partial
P5	Demo K	NEG · plateau ~800	RC3 PRIMARY
P7	Constant-total	diversity still wins	Partial
P9	FT/LoRA/LfS/froz	0.90/0.72/0.03/0.00	RC5 fold
P11	Recipe 32×50	SR 85-92.5%	RC6 Partial
P2	#objects N	Pour 0.133→0.939	RC2 PRIMARY
P4	Power-law fit	MA pairs $\alpha=-0.683$	RC4
P6	Objs/env heatmap	multi helps small M	Partial
P8	ViT-S/B/L	0.66/0.81/0.90	RC5
P10	U-Net S/B/L	0.88/0.90/0.83	RC5
P12	MSE power-law	weaker · $r=0.558$	N / light

RC0-RC5 smoke cards

Quiet = outlines only · do NOT sbatch · mini-N only

RC0

ingest

CPU <30m
env/obj/pair tags
mix-card schema

RC1

env M

$M \in \{1, 2, 4\}$
P1 PRIMARY
4-14 GPU-h

RC2

obj N

$N \in \{1, 2, 4\}$
P2 PRIMARY
4-14 GPU-h

RC3

demo K

$K \in \{10, 25, 50\}$
P5 PRIMARY
4-12 GPU-h

RC4

powerlaw

fit $Y = \beta X^\alpha$
 $\alpha < 0$ JSON
CPU <1 h

RC5

capacity

ViT-S vs B
U-Net S vs base
FT vs LoRA

NOT required: Tab.1 ~90% · Fig.5 $r > 0.94$ · M=32 · 1191 pairs · 8×A800 · 15k rollouts · 3 seeds · tester scores

Hypotheses (falsifiable) — mini-N

Success = directional ranking · NOT Tab.1 90% parity

H-env Env-OOD rises with $M \in \{1, 2, 4\}$ at fixed object (Pour 0.144→0.956 moral); env harder than object.

H-obj Object-OOD rises with $N \in \{1, 2, 4\}$ at fixed env (Pour 0.133→0.939).

H-demo-sat At fixed pairs, $K \in \{10, 25, 50\}$ shows diminishing returns; no strong power-law vs K .

H-power $Y=1-S$ vs $M/N/\text{pairs}$ is log-linear with $\alpha < 0$; demo-total alone weak.

H-vit Under matched mini-N, ViT-B > ViT-S on OOD (0.81 vs 0.66 moral).

H-unet-flat U-Net base $\nrightarrow$ small on OOD; large may hurt (0.88/0.90/0.83).

H-ft Full FT $\gg$ LoRA on stage proxy even if MSE ranks LoRA better.

H-div>vol At matched total demos, higher M or N beats low-diversity high- K .

H-recipe Mini high-pair $\times K \approx 50$ beats concentrated demos — without claiming Tab.1 90%.

Numbers card — memorize for exam

Study scale · exponents · plateaus · capacity · recipe

>40k / >15k

demos / Franka rollouts

-0.703 / -0.844 / -0.579 / 0.697 / -0.466 / -0.683

PW obj/env/pair α

MA obj/env/pair α

1191 pairs

MA $\rightarrow$ score 0.99

400 / 800 / 1600

plateaus @ 8/16/32 pairs

K = 50

recommended / pair

0.66 / 0.81 / 0.90

ViT-S / B / L

0.88 / 0.90 / 0.83

U-Net S / base / L

0.90 / 0.72 / 0.03 / 0

FT / LoRA / LfS / frozen

85-92.5%

Tab.1 SR · 4 tasks

~75 h / 8×A800

paper largest train

>396M

encoder + noise net

Motion — DP chunks + ACT temporal ensemble

@5 Hz Franka · action horizon 16 · obs horizon 2-3

DP

Diffusion Policy · CNN 1D U-Net

DDIM 16

inference denoising steps

ACT TE

exp-weighted · 8 steps · rate -0.01

Hor = 16

action horizon

5 Hz

env / control frequency

Obs 2-3

2 default; 3 Pour/Unplug (+0.25/0.5s)

224²

wrist RGB · GoPro fisheye

bs 256

AdamW · lr 3e-4 / 3e-5 enc

Not ACT-as-primary · not VLA · not classical planner · TE stitches DP chunks to reduce jerky transitions

Capacity — ViT scales; U-Net doesn't; FT required

Tab.2 · Pour Water · 32 pairs · 50% demos

Encoder strategy

Full FT DINOv2	0.90
LoRA rank=8	0.72
Learn-from-scratch	0.03
Frozen DINOv2	0.00

Visual encoder size

ViT-S/14	0.66
ViT-B/14	0.81
ViT-L/14	0.90
scales steadily	✓

Action U-Net size

small (512)	0.88
base	0.90
large (2048)	0.83
scale hurts	✗

Table 1 — recipe verification 32×50

8 unseen envs × 2 objs × 5 trials · ~90% SR · 4 collectors × 1 afternoon for 2 new tasks

Task	Norm. score	Success rate
Pour Water	0.922 ± 0.075	85.0 ± 19.4%
Mouse Arrangement	0.933 ± 0.088	92.5 ± 9.7%
Fold Towels	0.95 ± 0.062	87.5 ± 17.1%
Unplug Charger	0.887 ± 0.14	90.0 ± 14.1%

Per-env variance: Pour env#2 = 40% SR while others 80–100% (Tab.12) — product QA must stratify

Failure regimes — Empiricist must say

Demo-scaling fails past plateau · MSE $\neq$ reality · variance · DP-only

Demo-scaling fails

Past plateau, more hours on same (env,obj) $\neq$ more OOD. K=50 is enough for similar difficulty.

MSE $\neq$ reality

LoRA can win MSE (0.0049) yet lose score (0.72 vs 0.90). Never ship on val MSE alone.

Env harder than object

Shallower early slope when scaling M vs N for these tasks — log axes separately.

Per-env variance

Tab.12: Pour env#2 = 40% while mean 85%. Blind batch $\neq$ zero risk.

Batch incomparability

App.E.2: scores comparable within blind batch only — lock one eval shard.

DP-only / UMI confound

Exponents may shift for ACT/VLA/RL; UMI noise floor confounds “data law” vs “interface law.”

No task / lang axis

Cannot adjudicate generalist VLA scaling from these laws (§7).

UMI selection bias

Texture-poor / large-occluder scenes avoided — task-family bias.

GPU honesty — paper vs course

Paper 75 h / 8xA800 · course mini-N only · NEVER claim Tab.1 90% from smoke

Setup	GPU VRAM	Host RAM	Time
RC0 ingest	—	8-16 GB	<0.5 h CPU
RC1 env M×3	10-20 GB	32-64 GB	4-14 h total
RC2 obj N×3	10-20 GB	32-64 GB	4-14 h total
RC3 K×3	10-20 GB	32-64 GB	4-12 h total
RC4 power-law	—	8 GB	<1 h CPU
RC5 capacity×4	12-24 GB	64 GB	6-16 h total
Paper ViT-L @32×50	8xA800	large	~75 h largest
Paper 15k / 1191	robot+	—	OUT OF COURSE

Course envelope $\leq \sim 40\text{--}55$ GPU-h for P1+P2+P5+P4+P8/P10. Full grids $\times$ 4 tasks $\times$ real eval = hundreds-thousands — do not promise.

Figs. 2-4 — diversity curves (paper)

Object / env / pair generalization × demo fraction · 95% CI shaded

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mean squared error (MSE) on the validation set is another potential metric, we find it often does not correlate with real-world performance (see Appendix E.1 for more details). Finally, to minimize the tester's subjective bias, we simultaneously evaluate multiple policies trained on datasets of different sizes; each rollout is randomly selected from these multiple policies, while ensuring identical initial conditions for both the objects and the robot arm, enabling a fair comparison across policies. See Appendix E.2 for an example of the evaluation workflow and Appendix F for the hardware setup.

4 UNVEILING OF DATA SCALING LAWS

In this section, we first explore how increasing the number of training objects affects object generalization. Next, we analyze how the number of training environments impacts environment generalization. Finally, we study generalization across both dimensions simultaneously. Throughout all experiments, we also analyze the effect of demonstration quantity (Sec. 4.1). From these results, we derive the power-law data scaling laws (Sec. 4.2). Based on these laws, we further demonstrate an efficient data collection strategy to achieve a generalizable policy (Sec. 4.3).

4.1 RESULTS AND QUALITATIVE ANALYSIS

Tasks. We first focus on two manipulation tasks: *Pour Water* and *Mouse Arrangement*. In *Pour Water*, the robot performs three steps: first, it grabs a drinking bottle placed randomly on the table; second, it pours water into a mug; and finally, it places the bottle on a red coaster. This task demands precision, especially in aligning the bottle's mouth with the mug. In *Mouse Arrangement*, the robot completes two steps: it picks up a mouse and positions it on a mouse pad with its front facing forward. The mouse may be tilted, requiring the robot to employ non-prehensile actions (i.e., pushing) to first align it. Illustrations of all tasks are shown in Fig. 1, with further task details available in Appendix D. Robot rollout videos can be found on our website.

Object generalization. We use 32 distinct objects within the same environment to collect 120 demonstrations per object, yielding a total of 3,840 demonstrations for each task. After SLAM filtering, the number of valid demonstrations for *Pour Water* and *Mouse Arrangement* is reduced to 3,765 and 3,820, respectively. To investigate how the number of training objects influences the policy's ability to generalize to unseen objects, we randomly select 2^m objects ($m = 0, 1, 2, 3, 4, 5$) from the pool of 32 for training. Furthermore, to examine how policy performance varies with the number of demonstrations, we randomly sample 2^n fractions of valid demonstrations ($n = 0, -1, -2, -3, -4, -5$) for each selected object. For each combination of (m, n) , we train a policy if the total number of demonstrations exceeds 100. In total, 21 policies are trained, and each is evaluated using 8 unseen objects in the same environment as the training data, with 5 trials per object. The average normalized score across 40 trials is reported for each policy.

Fig. 2 presents the results, with shaded regions representing 95% confidence intervals. There are several key observations: (1) As the number of training objects increases, the policy's performance on unseen objects consistently improves across all fractions of demonstrations. (2) With more training objects, fewer demonstrations are required per object. For example, in *Pour Water*, when training with 8 objects, the performance using 12.5% of the demonstrations significantly lags behind that using 100% of the demonstrations; however, this gap nearly disappears when training with 32 objects. (3) Object generalization is relatively easy to achieve. The initial slope of the performance curve is very steep: with only 8 training objects, the normalized score for both tasks exceeds 0.8. When the number of training objects reaches 32, the score surpasses 0.9. These scores correspond to policies that have already generalized well to any new objects within the same category.

Environment generalization. To explore the effect of the number of training environments on generalization, we use the same manipulation object across 32 distinct environments, collecting 120 demonstrations per environment. For *Pour Water* and *Mouse Arrangement*, this result in 3,424 and 3,351 valid demonstrations, respectively. We randomly select 2^m environments ($m = 0, 1, 2, 3, 4, 5$) from the 32 available for training, and for each selected environment, we randomly select 2^n fractions of valid demonstrations ($n = 0, -1, -2, -3, -4, -5$). Each policy is evaluated in 8 unseen environments using the same object as in training, with 5 trials per environment.

Fig. 3 presents the results, revealing several notable patterns: (1) Increasing the number of training environments enhances the policy's generalization performance on unseen environments. This trend

Object gen “easy”; env harder; joint saturates demos faster — diversity $\gg$ depth

Numbers tattoo — pin these

Every deck carries the same spine numbers

diversity $\gg$ depth

envs/objs/pairs $\gg$ K

32 $\times$ 50

pairs $\times$ demos $\rightarrow$ $\sim$ 90%

85-92.5%

Tab.1 SR $\cdot$ 4 tasks

$\alpha \approx -0.47 \dots -0.84$

Fig.5 exponents

0.66 / 0.81 / 0.90

ViT-S / B / L

0.88 / 0.90 / 0.83

U-Net S / base / L

0.00 / 0.03

frozen / LfS fail

K $\approx$ 50

plateau / pair

>40k / >15k

demos / rollouts

1191 pairs

MA score 0.99 extrap.

DP + ACT TE

@5 Hz $\cdot$ horizon 16

FT 0.90 / LoRA 0.72

full FT required

Empiricist — close

Takeaways + exam bite

1. Fit $Y=1-S$ on $M/N/pairs$ — not on demo count past $K \approx 50$.
2. Primary smokes: RC1 env + RC2 obj + RC3 K; then RC4 fit + RC5 capacity.
3. GPU honesty: paper 75h/8×A800 out of course; mini-N only; never claim Tab.1 90% from smoke.
4. Tattoo: diversity $\gg$ depth · $32 \times 50 \rightarrow \sim 90\%$ · ViT scales / U-Net doesn't · frozen/LfS fail.

EXAM BITE

Lin et al. (ICLR'25 Oral): OOD gap $Y=1-S \approx \beta X^\alpha$ with $\alpha \sim -0.5 \dots -0.8$ on #envs/#objects/#pairs ($r \sim 0.94-0.99$); demos plateau ($\sim 50/pair$); recipe $32 \times 50 \rightarrow \sim 90\%$ ZS; diversity $\gg$ depth; encoder scale helps, U-Net doesn't.